Mohamed Rabie Ghediri

AI Engineering Student

+216 27 864 723 rabye.ghd@gmail.com Rabyegh Tunisia

Profile

Final-year Data Engineering and Decision Systems engineering student specializing in artificial intelligence, machine learning, NLP, computer vision, and generative AI. Experienced in designing AI-powered applications using LLMs, RAG, semantic search, forecasting, and intelligent retrieval systems. Seeking a final-year project internship in AI Engineering, Machine Learning, NLP, or Generative AI.

Experience

Yonnov'IA – AI / Data Science Intern Marseille, France
Python, FastAPI, React, TypeScript, PostgreSQL, Sentence Transformers, Statsmodels June – August 2026

- Built an end-to-end platform for scientific-domain discovery, historical publication analysis, and trend forecasting through 2030, supporting OpenAlex and local CSV datasets.
- Implemented E5 semantic discovery to construct research families and identify terminology variants from publication data.
- Developed a forecasting engine comparing Naive, Linear, Holt, ETS, and ARIMA models with rolling-origin backtesting (MAE, RMSE, MASE, sMAPE) for automatic model selection.

Otospex Solutions – Data Engineering Intern Djerba, Tunisia
Python, SQL, Airflow, dbt, PostgreSQL, Docker, Power BI June – August 2025

- Built an Airflow ELT pipeline loading multi-source vehicle data into a PostgreSQL warehouse.
- Built incremental dbt Bronze/Silver/Gold layers for clean, standardized star-schema data.
- Implemented automated data quality checks for completeness, validity, uniqueness, and consistency on vehicle listings.

Projects

RegulaRAG – AI-Powered Regulatory Intelligence & Compliance Analysis Platform
Hybrid RAG, GraphRAG, Elasticsearch, pgvector, Neo4j, FastAPI, React

Designed an AI-powered regulatory analysis platform combining Hybrid RAG and GraphRAG to retrieve, connect, and interpret complex regulatory information. Combined Elasticsearch and pgvector for hybrid retrieval, with Neo4j to represent relationships between regulatory entities and requirements. Developed the backend and user-facing application architecture using FastAPI and React.

AI Pharmaceutical Product Recognition & Price Comparison
YOLOv11, PaddleOCR, LLaMA 3.3, Groq, Elasticsearch, FastAPI, Vue

Developed a computer vision pipeline to recognize pharmaceutical products from images using YOLOv11 and extract product information with PaddleOCR. Integrated LLaMA 3.3 through Groq to process extracted information and support product comparison, with Elasticsearch for information retrieval. Achieved **92% recognition accuracy** with an average processing time of **3.2 seconds**.

CropWatch – Geospatial AI & Agricultural Intelligence Platform
Python, LightGBM, Scikit-learn, MLflow, Sentinel-2, CHIRPS, ERA5-Land, SoilGrids, Qdrant, RAG, LangGraph, FastAPI

Designing a regional agricultural monitoring system that combines satellite imagery, rainfall, climate, and soil indicators to assess cereal vegetation conditions. Developing a geospatial machine learning approach to identify regional drought-stress patterns and estimate the risk of vegetation-condition deterioration over a **14-day horizon**. Planning an explainable intelligence layer using retrieval-augmented generation to provide evidence-grounded insights from geospatial indicators and agricultural data.

Technical Skills

Programming Languages: Python, SQL, Java, Scala, JavaScript/TypeScript
AI & Generative AI: LLMs, RAG, NLP, Semantic Search, Hugging Face, LangChain, LangGraph, FAISS
Machine Learning & Deep Learning: Scikit-learn, PyTorch, TensorFlow, MLflow
Data Engineering: Apache Kafka, Apache Spark, Apache Airflow, Databricks, Hadoop, Docker, ETL/ELT Pipelines
Databases & Search: PostgreSQL, MongoDB, Cassandra, Elasticsearch, Vector Databases
Frameworks & Tools: FastAPI, React, Vue.js, Git, Power BI
Cloud & Deployment: AWS S3, Docker, Kubernetes, GitHub Actions
Languages: Arabic (Native), English (Advanced), French (Intermediate), German (Beginner)

Education

Engineering Degree – Data Engineering & Decision Systems (IDSD) 2024 – Present
ENET'Com, Sfax

Engineering Preparatory Cycle – Physics & Technology (PT) 2021–2024
IPEIN, Nabeul